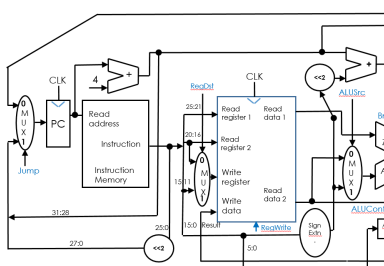
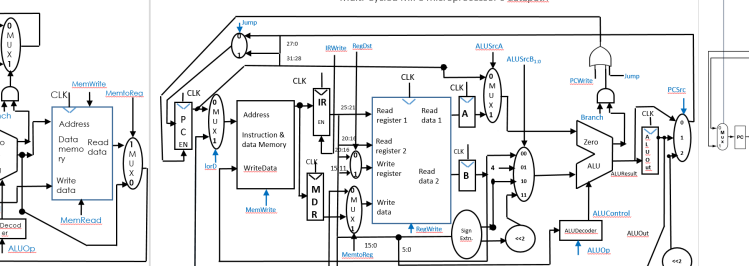


Single-Cycle MIPS microprocessor's datapath



Single-cycle (exam fig): M1 Jump (left), M2 RegDst, M3 ALUSrc, M4 PCSrc (top), M5 MemtoReg, A1 = PC+4, A2 = branch adder.

Multi-Cycle MIPS microprocessor's datapath



Multicycle (exam fig): M0 IorD, M1 Jump, M2 RegDst, M3 MemtoReg, M4 ALUSrcA, M5 ALUSrcB, M6 PCSrc; non-arch regs: IR, MDR, A, B, ALUOut.

Pipelined datapath + control (slide 32; PS&H; control in ID → WB/EX fields in ID/EX [9b], EX/EXMEM (5b), MEM/WB (2b).

type	31:26	25:21	20:16	15:11	10:6	5:0
PC	op(6)→PC	rs	rt	rd	shamt	funct
op	rs	rt	rd	shamt	funct	
rs	rt	rd	shamt	funct		
rt	rd	shamt	funct			
rd	shamt	funct				
shamt	funct					
funct						

1. Register: all operands in regs → add \$0,\$s1,\$s2
2. Immediate: 16-bit constant in inst (sign-ext) → add \$0,\$s1,\$s2
3. Base (displacement): add → R[rs] + SignExt(imm) → lw \$t0,\$s1(\$t0)
4. PCrelative: target = PC+4 + SignExt(imm) → beq/bne
5. Pseudo-direct: target = (PC+31:28), add26, 00] → j, jal

Design principles: simplicity favors regularity; make the common case fast; smaller is faster; good design demands good compromises. MIPS good for pipelining (slide 34): all instrs 32-bit, few addressing modes & formats, memory operands only in lw/sw, operands aligned (x86 1-15 bytes → hard).

Inst	op	funcn	Operation
add	000000	100000 (20)	rd←rs+rt
addi	000001	100010 (22)	rd←rs+imm
and	000010	100101 (22)	rd←rs&rt
andi	000011	100110 (22)	rd←rs&imm
beq	000100	101001 (20)	if(rs==rt) jump
bne	000101	101010 (20)	if(rs!=rt) jump
lwr	000110	101101 (20)	PC←rs
lwr	000111	101110 (20)	PC←rs+imm
lwr	000112	101111 (20)	PC←rs+imm
lwr	000113	101112 (20)	PC←rs+imm
lwr	000114	101113 (20)	PC←rs+imm
lwr	000115	101114 (20)	PC←rs+imm
lwr	000116	101115 (20)	PC←rs+imm
lwr	000117	101116 (20)	PC←rs+imm
lwr	000118	101117 (20)	PC←rs+imm
lwr	000119	101118 (20)	PC←rs+imm
lwr	000120	101119 (20)	PC←rs+imm
lwr	000121	101120 (20)	PC←rs+imm
lwr	000122	101121 (20)	PC←rs+imm
lwr	000123	101122 (20)	PC←rs+imm
lwr	000124	101123 (20)	PC←rs+imm
lwr	000125	101124 (20)	PC←rs+imm
lwr	000126	101125 (20)	PC←rs+imm
lwr	000127	101126 (20)	PC←rs+imm
lwr	000128	101127 (20)	PC←rs+imm
lwr	000129	101128 (20)	PC←rs+imm
lwr	000130	101129 (20)	PC←rs+imm
lwr	000131	101130 (20)	PC←rs+imm
lwr	000132	101131 (20)	PC←rs+imm
lwr	000133	101132 (20)	PC←rs+imm
lwr	000134	101133 (20)	PC←rs+imm
lwr	000135	101134 (20)	PC←rs+imm
lwr	000136	101135 (20)	PC←rs+imm
lwr	000137	101136 (20)	PC←rs+imm
lwr	000138	101137 (20)	PC←rs+imm
lwr	000139	101138 (20)	PC←rs+imm
lwr	000140	101139 (20)	PC←rs+imm
lwr	000141	101140 (20)	PC←rs+imm
lwr	000142	101141 (20)	PC←rs+imm
lwr	000143	101142 (20)	PC←rs+imm
lwr	000144	101143 (20)	PC←rs+imm
lwr	000145	101144 (20)	PC←rs+imm
lwr	000146	101145 (20)	PC←rs+imm
lwr	000147	101146 (20)	PC←rs+imm
lwr	000148	101147 (20)	PC←rs+imm
lwr	000149	101148 (20)	PC←rs+imm
lwr	000150	101149 (20)	PC←rs+imm
lwr	000151	101150 (20)	PC←rs+imm
lwr	000152	101151 (20)	PC←rs+imm
lwr	000153	101152 (20)	PC←rs+imm
lwr	000154	101153 (20)	PC←rs+imm
lwr	000155	101154 (20)	PC←rs+imm
lwr	000156	101155 (20)	PC←rs+imm
lwr	000157	101156 (20)	PC←rs+imm
lwr	000158	101157 (20)	PC←rs+imm
lwr	000159	101158 (20)	PC←rs+imm
lwr	000160	101159 (20)	PC←rs+imm
lwr	000161	101160 (20)	PC←rs+imm
lwr	000162	101161 (20)	PC←rs+imm
lwr	000163	101162 (20)	PC←rs+imm
lwr	000164	101163 (20)	PC←rs+imm
lwr	000165	101164 (20)	PC←rs+imm
lwr	000166	101165 (20)	PC←rs+imm
lwr	000167	101166 (20)	PC←rs+imm
lwr	000168	101167 (20)	PC←rs+imm
lwr	000169	101168 (20)	PC←rs+imm
lwr	000170	101169 (20)	PC←rs+imm
lwr	000171	101170 (20)	PC←rs+imm
lwr	000172	101171 (20)	PC←rs+imm
lwr	000173	101172 (20)	PC←rs+imm
lwr	000174	101173 (20)	PC←rs+imm
lwr	000175	101174 (20)	PC←rs+imm
lwr	000176	101175 (20)	PC←rs+imm
lwr	000177	101176 (20)	PC←rs+imm
lwr	000178	101177 (20)	PC←rs+imm
lwr	000179	101178 (20)	PC←rs+imm
lwr	000180	101179 (20)	PC←rs+imm
lwr	000181	101180 (20)	PC←rs+imm
lwr	000182	101181 (20)	PC←rs+imm
lwr	000183	101182 (20)	PC←rs+imm
lwr	000184	101183 (20)	PC←rs+imm
lwr	000185	101184 (20)	PC←rs+imm
lwr	000186	101185 (20)	PC←rs+imm
lwr	000187	101186 (20)	PC←rs+imm
lwr	000188	101187 (20)	PC←rs+imm
lwr	000189	101188 (20)	PC←rs+imm
lwr	000190	101189 (20)	PC←rs+imm
lwr	000191	101190 (20)	PC←rs+imm
lwr	000192	101191 (20)	PC←rs+imm
lwr	000193	101192 (20)	PC←rs+imm
lwr	000194	101193 (20)	PC←rs+imm
lwr	000195	101194 (20)	PC←rs+imm
lwr	000196	101195 (20)	PC←rs+imm
lwr	000197	101196 (20)	PC←rs+imm
lwr	000198	101197 (20)	PC←rs+imm
lwr	000199	101198 (20)	PC←rs+imm
lwr	000200	101199 (20)	PC←rs+imm
lwr	000201	101200 (20)	PC←rs+imm
lwr	000202	101201 (20)	PC←rs+imm
lwr	000203	101202 (20)	PC←rs+imm
lwr	000204	101203 (20)	PC←rs+imm
lwr	000205	101204 (20)	PC←rs+imm
lwr	000206	101205 (20)	PC←rs+imm
lwr	000207	101206 (20)	PC←rs+imm
lwr	000208	101207 (20)	PC←rs+imm
lwr	000209	101208 (20)	PC←rs+imm
lwr	000210	101209 (20)	PC←rs+imm
lwr	000211	101210 (20)	PC←rs+imm
lwr	000212	101211 (20)	PC←rs+imm
lwr	000213	101212 (20)	PC←rs+imm
lwr	000214	101213 (20)	PC←rs+imm
lwr	000215	101214 (20)	PC←rs+imm
lwr	000216	101215 (20)	PC←rs+imm
lwr	000217	101216 (20)	PC←rs+imm
lwr	000218	101217 (20)	PC←rs+imm
lwr	000219	101218 (20)	PC←rs+imm
lwr	000220	101219 (20)	PC←rs+imm
lwr	000221	101220 (20)	PC←rs+imm
lwr	000222	101221 (20)	PC←rs+imm
lwr	000223	101222 (20)	PC←rs+imm
lwr	000224	101223 (20)	PC←rs+imm
lwr	000225	101224 (20)	PC←rs+imm
lwr	000226	101225 (20)	PC←rs+imm
lwr	000227	101226 (20)	PC←rs+imm
lwr	000228	101227 (20)	PC←rs+imm
lwr	000229	101228 (20)	PC←rs+imm
lwr	000230	101229 (20)	PC←rs+imm
lwr	000231	101230 (20)	PC←rs+imm
lwr	000232	101231 (20)	PC←rs+imm
lwr	000233	101232 (20)	PC←rs+imm
lwr	000234	101233 (20)	PC←rs+imm
lwr	000235	101234 (20)	PC←rs+imm
lwr	000236	101235 (20)	PC←rs+imm
lwr	000237	101236 (20)	PC←rs+imm
lwr	000238	101237 (20)	PC←rs+imm
lwr	000239	101238 (20)	PC←rs+imm
lwr	000240	101239 (20)	PC←rs+imm
lwr	000241	101240 (20)	PC←rs+imm
lwr	000242	101241 (20)	PC←rs+imm
lwr	000243	101242 (20)	PC←rs+imm
lwr	000244	101243 (20)	PC←rs+imm
lwr	000245	101244 (20)	PC←rs+imm
lwr	000246	101245 (20)	PC←rs+imm
lwr	000247	101246 (20)	PC←rs+imm
lwr	000248	101247 (20)	PC←rs+imm
lwr	000249	101248 (20)	PC←rs+imm
lwr	000250	101249 (20)	PC←rs+imm
lwr	000251	101250 (20)	PC←rs+imm
lwr	000252	101251 (20)	PC←rs+imm
lwr	000253	101252 (20)	PC←rs+imm
lwr	000254	101253 (20)	PC←rs+imm
lwr	000255	101254 (20)	PC←rs+imm
lwr	000256	101255 (20)	PC←rs+imm
lwr	000257	101256 (20)	PC←rs+imm
lwr	000258	101257 (20)	PC←rs+imm
lwr	000259	101258 (20)	PC←rs+imm
lwr	000260	101259 (20)	PC←rs+imm
lwr	000261	101260 (20)	PC←rs+imm
lwr	000262	101261 (20)	PC←rs+imm
lwr	000263	101262 (20)	PC←rs+imm
lwr	000264	101263 (20)	PC←rs+imm
lwr	000265	101264 (20)	PC←rs+imm
lwr	000266	101265 (20)	PC←rs+imm
lwr	000267	101266 (20)	PC←rs+imm
lwr	000268	101267 (20)	PC←rs+imm
lwr	000269	101268 (20)	PC←rs+imm
lwr	000270	101269 (20)	PC←rs+imm
lwr	000271	101270 (20)	PC←rs+imm
lwr	000272	101271 (20)	PC←rs+imm
lwr	000273	101272 (20)	PC←rs+imm
lwr	000274	101273 (20)	PC←rs+imm
lwr	000275	101274 (20)	PC←rs+imm
lwr	000276	101275 (20)	PC←rs+imm
lwr	000277	101276 (20)	PC←rs+imm
lwr	000278	101277 (20)	PC←rs+imm
lwr	000279	101278 (20)	PC←rs+imm
lwr	000280	101279 (20)	PC←rs+imm
lwr	000281	101280 (20)	PC←rs+imm
lwr	000282	101281 (20)	PC←rs+imm
lwr	000283	101282 (20)	PC←rs+imm
lwr	000284	101283 (20)	PC←rs+imm
lwr	000285	101284 (20)	PC←rs+imm
lwr	000286	101285 (20)	PC←rs+imm
lwr	000287	101286 (20)	PC←rs+imm
lwr	000288	101287 (20)	PC←rs+imm
lwr	000289	101288 (20)	PC←rs+imm
lwr	000290	101289 (20)	PC←rs+imm
lwr	000291	101290 (20)	PC←rs+imm
lwr	000292	101291 (20)	PC←rs+imm
lwr	000293	101292 (20)	PC←rs+imm
lwr	000294	101293 (20)	PC←rs+imm
lwr	000295	101294 (20)	PC←rs+imm
lwr	000296	101295 (20)	PC←rs+imm
lwr	000297	101296 (20)	PC←rs+imm
lwr	000298	101297 (20)	PC←rs+imm
lwr	000299	101298 (20)	PC←rs+imm
lwr	000300	101299 (20)	PC←rs+imm
lwr	000301	101300 (20)	PC←rs+imm
lwr	000302	101301 (20)	PC←rs+imm
lwr	000303	101302 (20)	PC←rs+imm
lwr	000304	101303 (20)	PC←rs+imm
lwr	000305	101304 (20)	PC←rs+imm
lwr	000306	101305 (20)	PC←rs+imm
lwr	000307	101306 (20)	PC←rs+imm
lwr	000308	101307 (20)	PC←rs+imm
lwr	000309	101308 (20)	PC←rs+imm
lwr	000310	101309 (20)	PC←rs+imm
lwr	000311	101310 (20)	PC←rs+imm
lwr	000312	101311 (20)	PC←rs+imm
lwr	000313	101312 (20)	PC←rs+imm
lwr	000314	101313 (20)	PC←rs+imm
lwr	000315	101314 (20)	PC←rs+imm
lwr	000316	101315 (20)	PC←rs+imm
lwr	000317	101316 (20)	PC←rs+imm
lwr	000318	101317 (20)	PC←rs+imm
lwr	000319	101318 (20)	PC←rs+imm
lwr	000320	101319 (20)	PC←rs+imm
lwr	000321	101320 (20)	PC←rs+imm
lwr	000322	101321 (20)	PC←rs+imm
lwr	000323	101322 (20)	PC←rs+imm
lwr	000324	101323 (20)	PC←rs+imm
lwr	000325	101324 (20)	PC←rs+imm
lwr	000326	101325 (20)	PC←rs+imm
lwr	000327	101326 (20)	PC←rs+imm
lwr	000328	101327 (20)	PC←rs+imm
lwr	000329	101328 (20)	PC←rs+